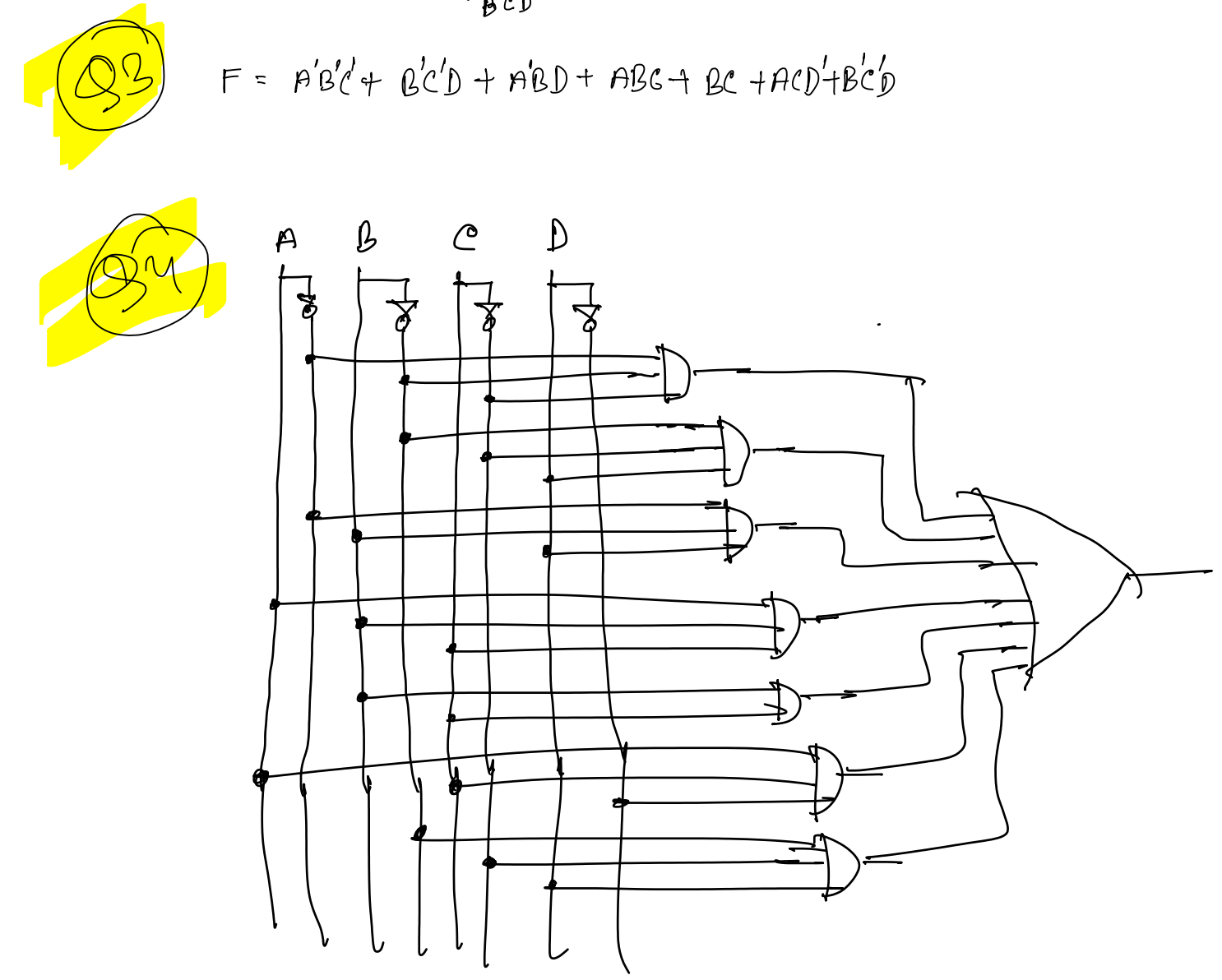
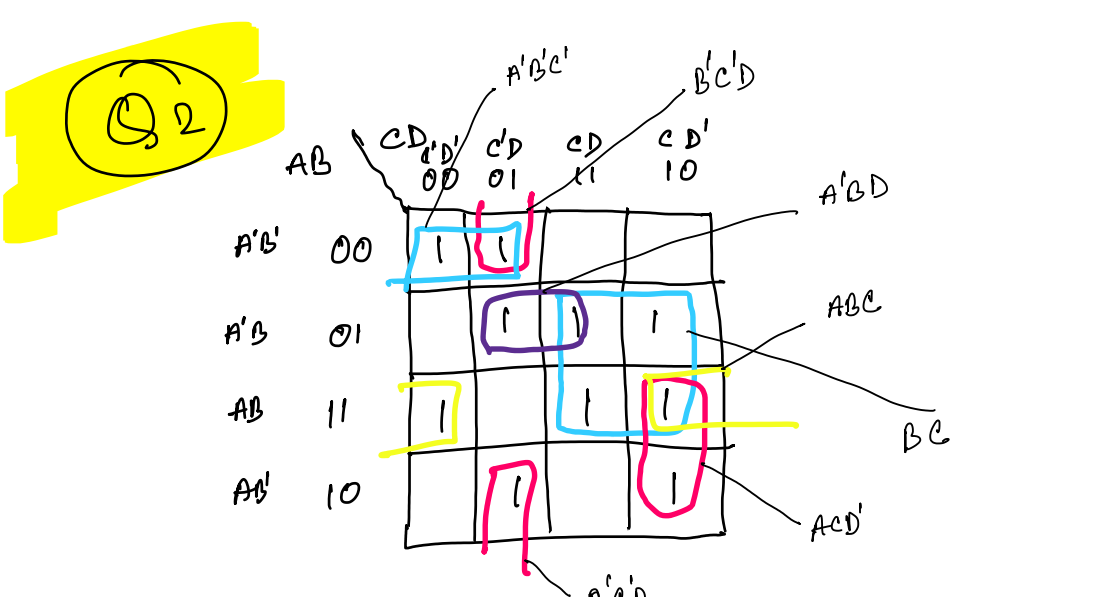


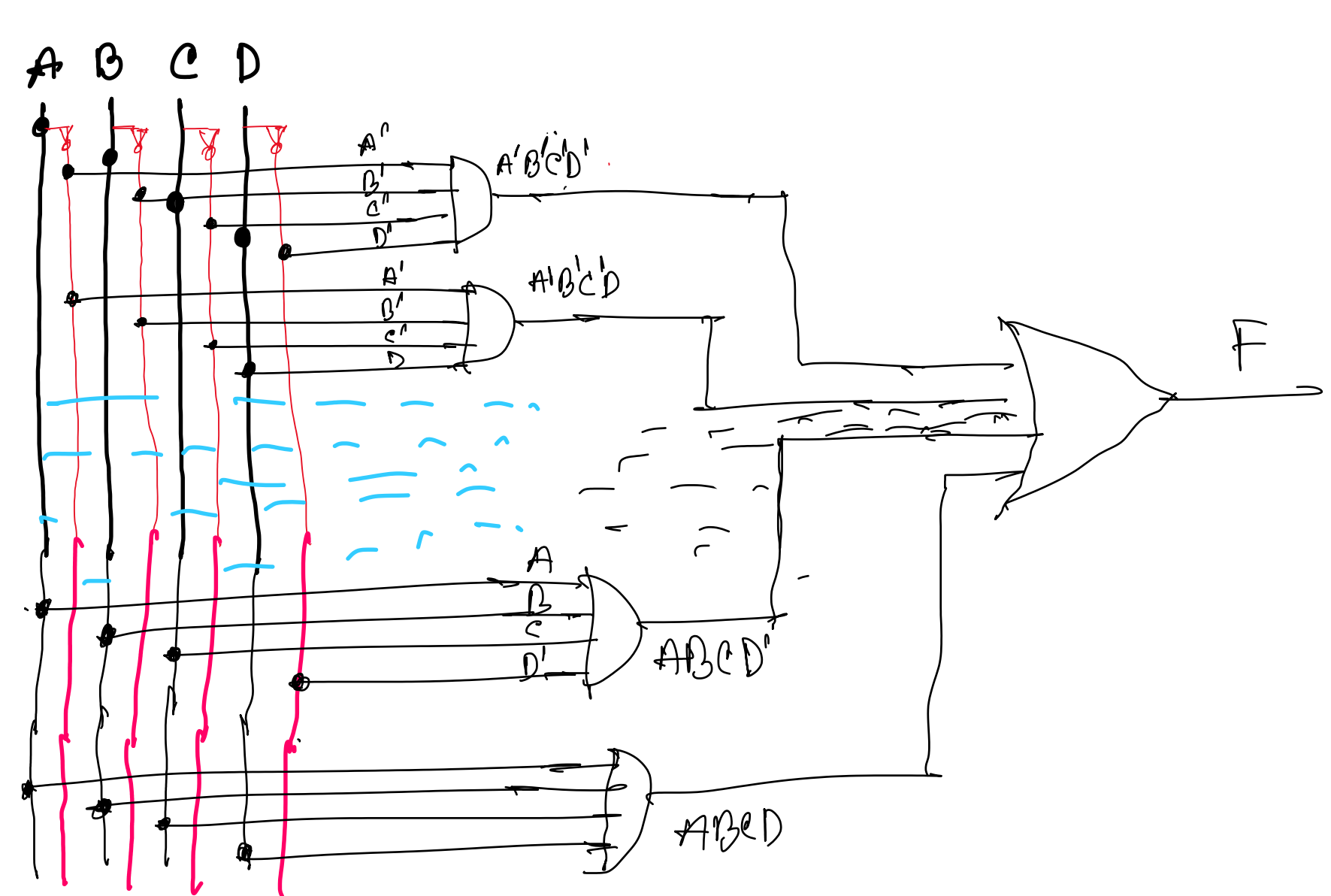
Set B

$F(A,B,C,D) = \sum(0,1,5,6,7,9,10,12,14,15)$

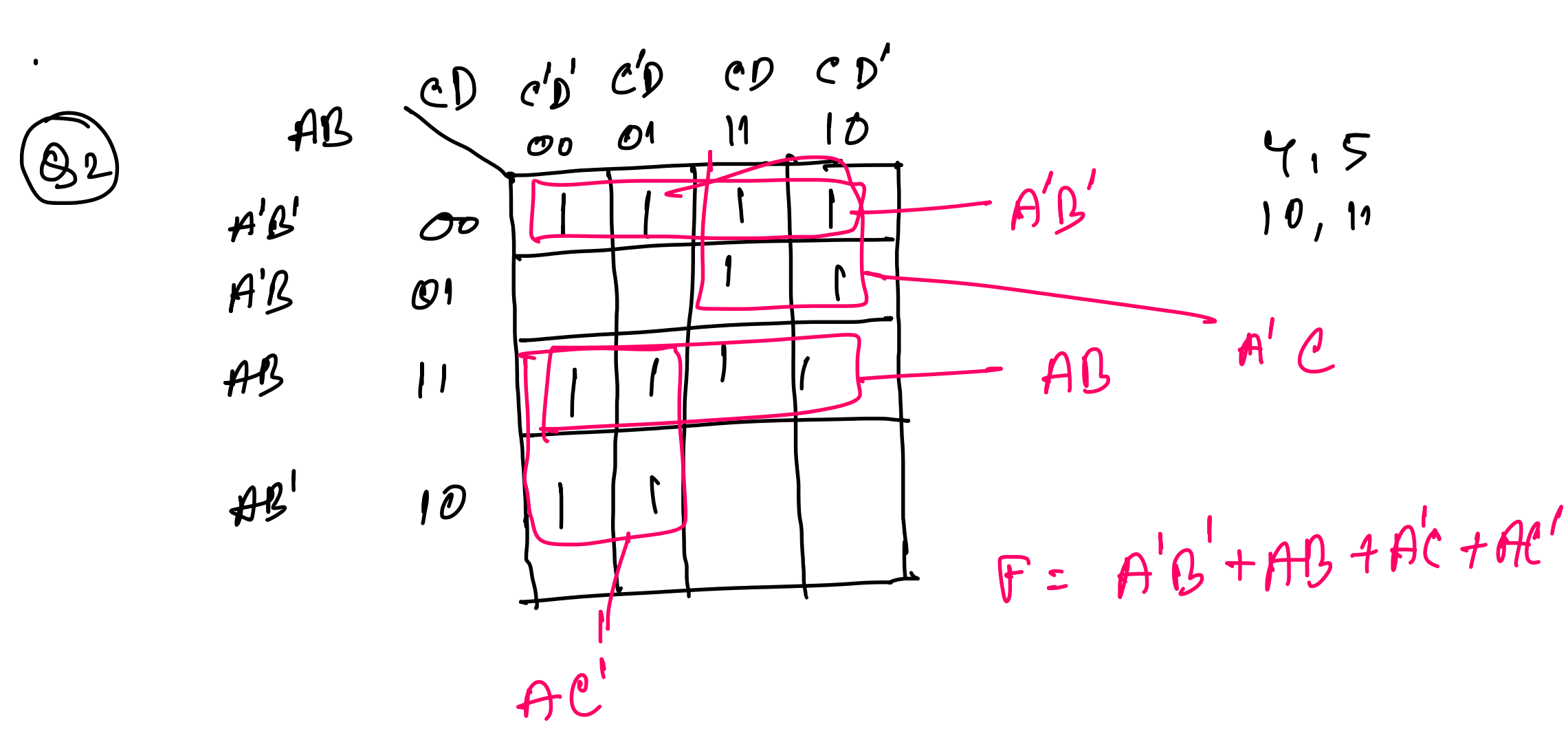
$= (A'B'C'D' + A'B'C'D + A'BC'D + A'BC'D' + A'BCD + AB'C'D + AB'C'D' + ABC'D + ABCD)$



In terms of Complexity Cost Time
Q4 - Circuit is better than Q1



Set A



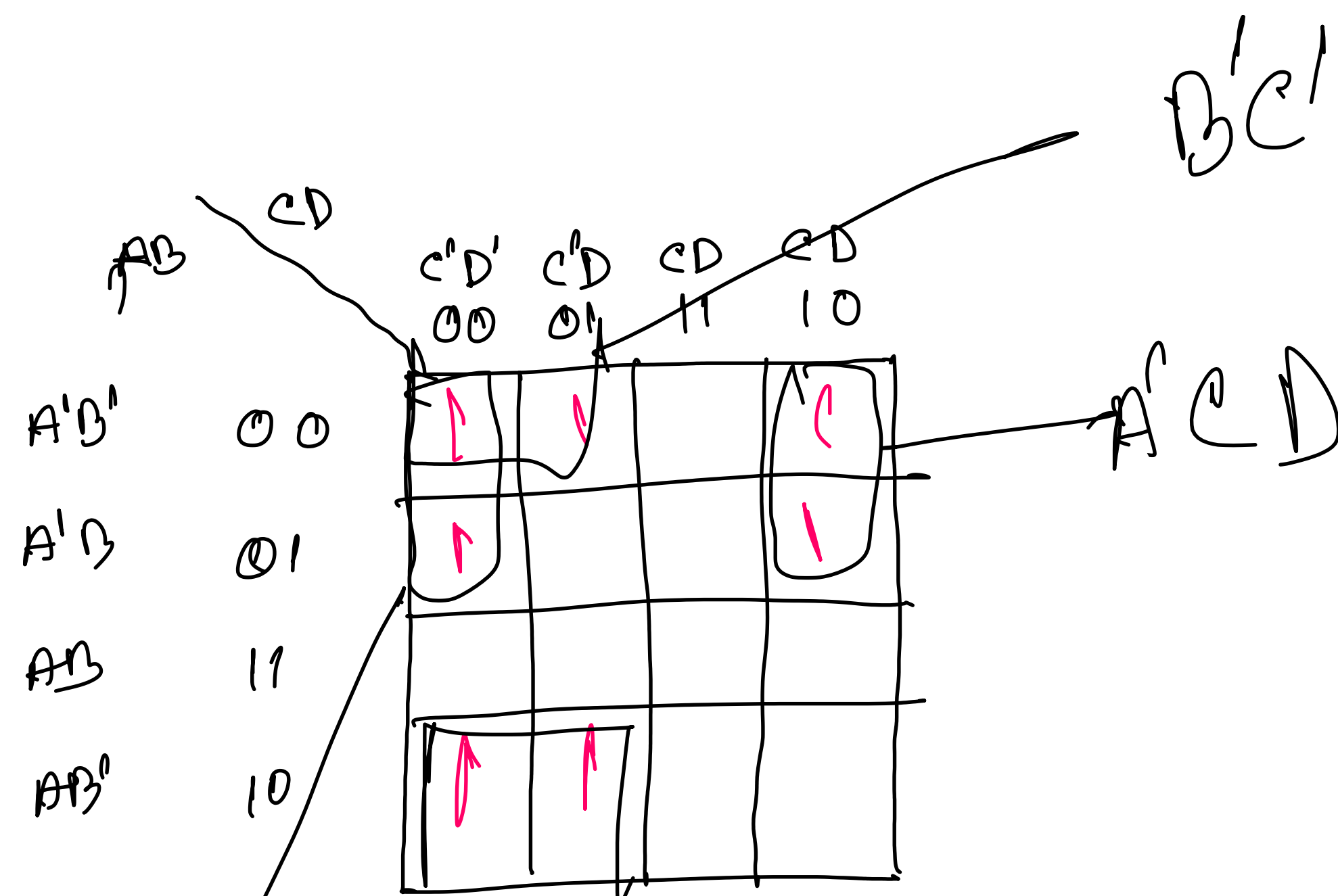
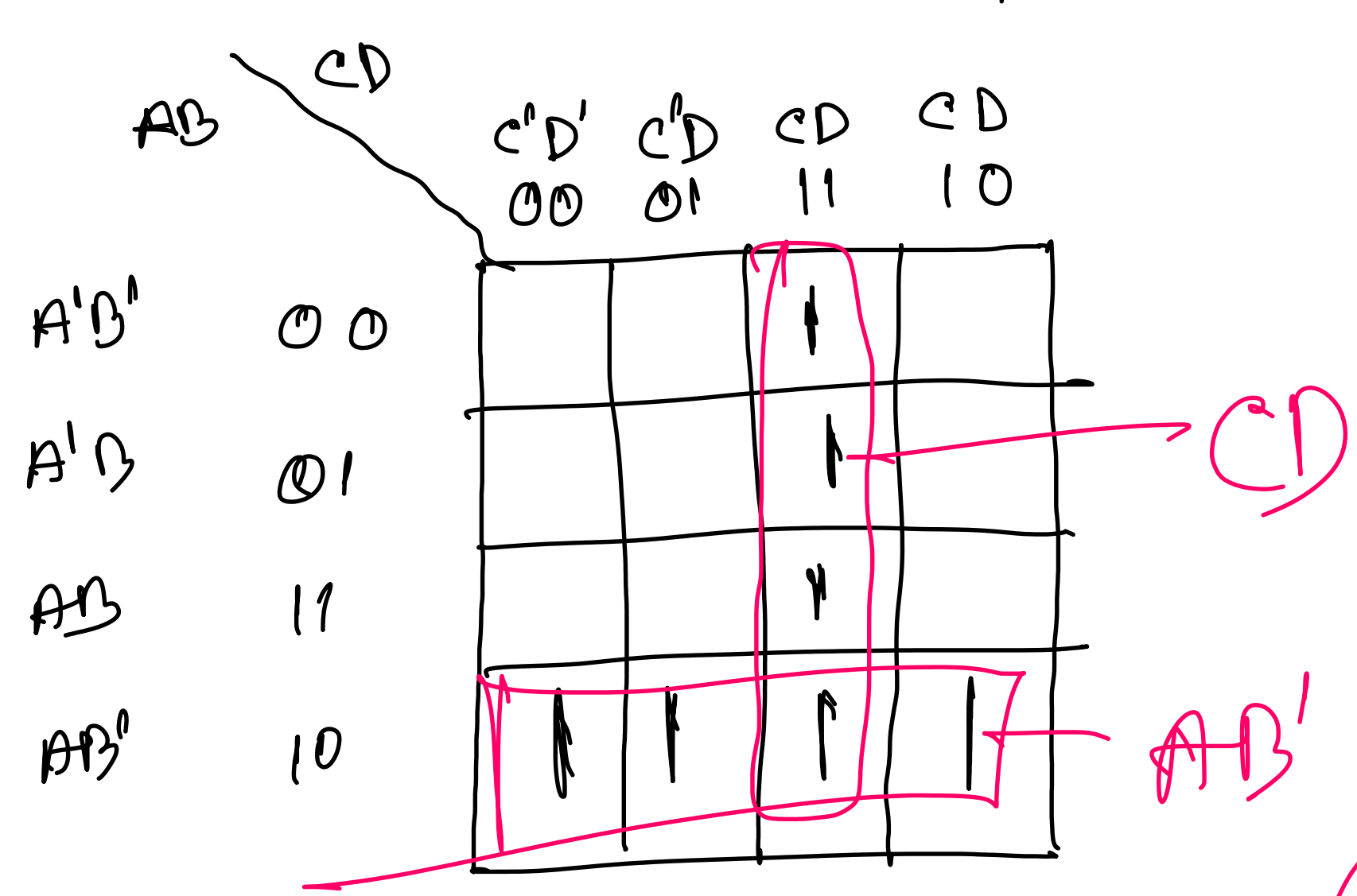
* Other questions answer structure will be same as Set B

Set B (P2)

	A	B	C	D	M	N
0	0	0	0	0	0	1
1	0	0	0	1	0	1
2	0	0	1	0	0	1
3	0	0	1	1	1	0
4	0	1	0	0	0	1
5	0	1	0	1	0	0
6	0	1	1	0	0	1
7	0	1	1	1	1	0
8	1	0	0	0	1	1
9	1	0	0	1	1	1
10	1	0	1	0	1	0
11	1	0	1	1	1	0
12	1	1	0	0	0	0
13	1	1	0	1	0	0
14	1	1	1	0	0	0
15	1	1	1	1	0	0

For M
A=1 & B=0
OR
C=1 & D=1

For N
D=0 & A=0
OR
B=0 & C=0



Set A (Problem)

	A	B	C	D	M	N
0	0	0	0	0	0	1
1	0	0	0	1	0	1
2	0	0	1	0	1	1
3	0	0	1	1	0	1
4	0	1	0	0	0	1
5	0	1	0	1	0	0
6	0	1	1	0	1	0
7	0	1	1	1	0	1
8	1	0	0	0	0	0
9	1	0	0	1	0	0
10	1	0	1	0	1	0
11	1	0	1	1	0	1
12	1	1	0	0	1	0
13	1	1	0	1	1	0
14	1	1	1	0	1	0
15	1	1	1	1	1	0

M
C=1 & D=0
OR
A=1 & B=1

N
B=0 & A=0
OR
C=1 & D=1

